

Courts in Demand

Do Tennis Clinics Crowd Out Regular Play at High Hampton Country Club?

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Introduction

High Hampton Country Club operates five tennis courts: four Har-Tru clay courts and one hard court. During the summer, the club's peak season, these courts serve both regular member and inn-guest reservations and organized programming such as tennis clinics, lessons, and court bookings. This creates a practical capacity constraint: during high-demand periods, the club may not have enough available courts to accommodate every member or inn guest seeking a reservation while also operating scheduled clinics. Based on years of experience working at the club, periods of simultaneous demand were common during the summer, with members and guests sometimes seeking court access while clinic programming was underway. Understanding how clinic scheduling relates to regular court use is therefore important for balancing programming opportunities with reliable access to courts.

This analysis investigates whether tennis clinics are associated with changes in daily member and inn-guest court bookings across four summer seasons. Specifically, it considers whether clinic programming appears to reduce regular court-booking access and whether the scheduling of clinics represents a reasonable balance between organized programming and member access to courts.

The data available for this analysis measure daily booking activity and clinic participation, rather than court-hours, specific clinic times, court capacity, or financial revenue. Therefore, this paper focuses on the portion of the question that can be evaluated directly: whether days with clinics, particularly days with larger clinics, are associated with lower volumes of regular member and inn-guest court bookings.

At first glance, clinics might be expected to reduce ordinary bookings because courts are a finite resource. If a clinic uses one or more courts during periods when members would otherwise reserve them, a negative relationship between clinic participation and ordinary bookings may emerge. Alternatively, clinics may increase regular court use by helping newer or less experienced members develop the skills and confidence needed to play independently. Members who participate in a clinic may later be more likely to reserve a court for their own practice or play with other members and guests.

These competing possibilities led to the following research question: Do tennis clinics reduce daily member and inn-guest court bookings at High Hampton Country Club, or are clinics associated with greater overall court use?

Data

The analysis draws on daily tennis activity records from High Hampton Country Club across four summer seasons: 2023 through 2026. Each workbook documents court use during the club's busiest months, June, July, and August, providing a day-by-day view of regular bookings and clinic participation across the summer season.

Each observation represents one calendar day. The primary outcome variable is tennis bookings, defined as the sum of member court bookings and inn-guest court bookings:

Tennis bookings = Member court bookings + Inn Guest court bookings.

The primary explanatory variable is tennis clinic participation, defined as the sum of member and inn-guest clinic participants:

Tennis clinic participation = Member clinic participants + Inn Guest clinic participants

The analysis also includes several calendar variables that could influence ordinary tennis demand independently of clinics:

- Weekday: Monday through Sunday.
- Month: June, July, or August.
- Year: 2023, 2024, 2025, or 2026.
- Clinic size category: No clinic, 1–5 participants, 6–10 participants, or 11 or more participants.

The weekday, month, and year controls are important because tennis activity is not evenly distributed across the summer. For example, bookings may naturally be higher on weekends, during July, or in particular years. Without accounting for these factors, a simple comparison of clinic and non-clinic days could incorrectly attribute predictable calendar demand to clinic programming.

Below, Table 1 provides an illustrative preview of the daily analytical dataset. Each row represents one day of tennis activity and includes regular member and inn-guest bookings, clinic participation, and calendar characteristics.

Table 1. Excerpt of the Daily Tennis Activity Dataset

Date	Year	Weekday	Month	Member bookings	Inn-guest bookings	Tennis bookings	Clinic participants	Clinic size
2023								
2023-06-07	2023	Wednesday	June	7	2	9	0	No clinic
2023-07-22	2023	Saturday	July	15	7	22	8	6–10
2024								
2024-06-15	2024	Saturday	June	11	4	15	5	1–5
2024-07-27	2024	Saturday	July	17	8	25	11	11+
2025								
2025-06-10	2025	Tuesday	June	8	3	11	2	1–5
2025-08-02	2025	Saturday	August	16	6	22	7	6–10
2026								
2026-06-13	2026	Saturday	June	10	4	14	6	6–10
2026-07-18	2026	Saturday	July	14	7	21	12	11+

The cleaned dataset included 351 observations across 4 summer seasons, spanning June 1st, 2023 through August 14, 2026. Across all observed days, High Hampton Country Club recorded 3,273 regular member and inn-guest court bookings, averaging about 9.3 bookings per day. Tennis clinics were held on 175 days, representing 49.9% of the dataset, while 176 observed days had no clinic. Across clinic days, the club recorded 1,155 total clinic participants. The average clinic included 6.6 participants.

Modeling Approach

To evaluate the relationship between tennis clinics and regular court access, I modeled the number of daily member and inn-guest tennis court bookings. Because the outcome variable is a daily count, a standard linear regression model would not be ideal: it can produce unrealistic negative predictions and does not naturally account for the uneven variability commonly seen in count data. A Poisson regression model is designed for count outcomes, but it assumes that the average count and variability in the count are equal. The booking data showed greater variation than this assumption allows, so I used negative-binomial regression instead. Negative-binomial regression is designed for count data with this additional variation, or overdispersion.

The primary model estimated the association between the number of clinic participants and the expected number of regular court bookings on the same day. The model included weekday, month, and year as calendar controls because court demand may vary predictably across the week, throughout the summer season, and between years. For example, Saturdays may have more booking demand than Mondays, and July may have more tennis activity than

June regardless of whether a clinic is scheduled. Including these variables helps separate the relationship between clinics and bookings from normal calendar-based changes in demand.

The primary model was:

$$\log(E[\text{Tennis bookings}]_i) = B_0 + B_1(\text{Clinic Participants}_i) + B_2(\text{Weekday}_i) + B_3(\text{Month}_i) + B_4(\text{Year}_i)$$

In this model, i represents one calendar day. The dependent variable is the number of regular member and inn-guest court bookings on that day. The main predictor is the total number of tennis clinic participants. Weekday, month, and year are categorical variables, allowing the model to account for recurring demand patterns without assuming that demand changes in a simple straight-line pattern across days, months, or seasons.

Because negative-binomial coefficients are estimated on a logarithmic scale, I converted the model coefficients into booking rate ratios by exponentiating them. A rate ratio above 1.00 indicates higher expected booking volume, while a rate ratio below 1.00 indicates lower expected booking volume. For the clinic-participation coefficient, the rate ratio represents the proportional change in expected daily bookings associated with one additional clinic participant, holding weekday, month, and year constant.

I also estimated a second model that replaced the continuous clinic-participant variable with clinic-size categories: no clinic, 1–5 participants, 6–10 participants, and 11 or more participants. This model tests whether the relationship between clinic participation and bookings changes at different clinic sizes rather than assuming that each additional participant has the exact same relationship with bookings. No-clinic days served as the reference group. The category model provides a more operationally intuitive comparison between days without clinics and days with small, medium, or larger clinics.

The secondary model was:

$$\log(E[\text{Tennis bookings}]_i) = B_0 + B_1(\text{Clinic Size}) + B_2(\text{Weekday}_i) + B_3(\text{Month}_i) + B_4(\text{Year}_i)$$

The clinic-participant model served as the primary model because it uses the full range of observed clinic sizes and requires fewer estimated parameters. The clinic-size-category model was used as a sensitivity analysis to determine whether the results were consistent when clinic activity was measured in operationally meaningful groups. Both models adjust for the same weekday, month, and year variables.

Results

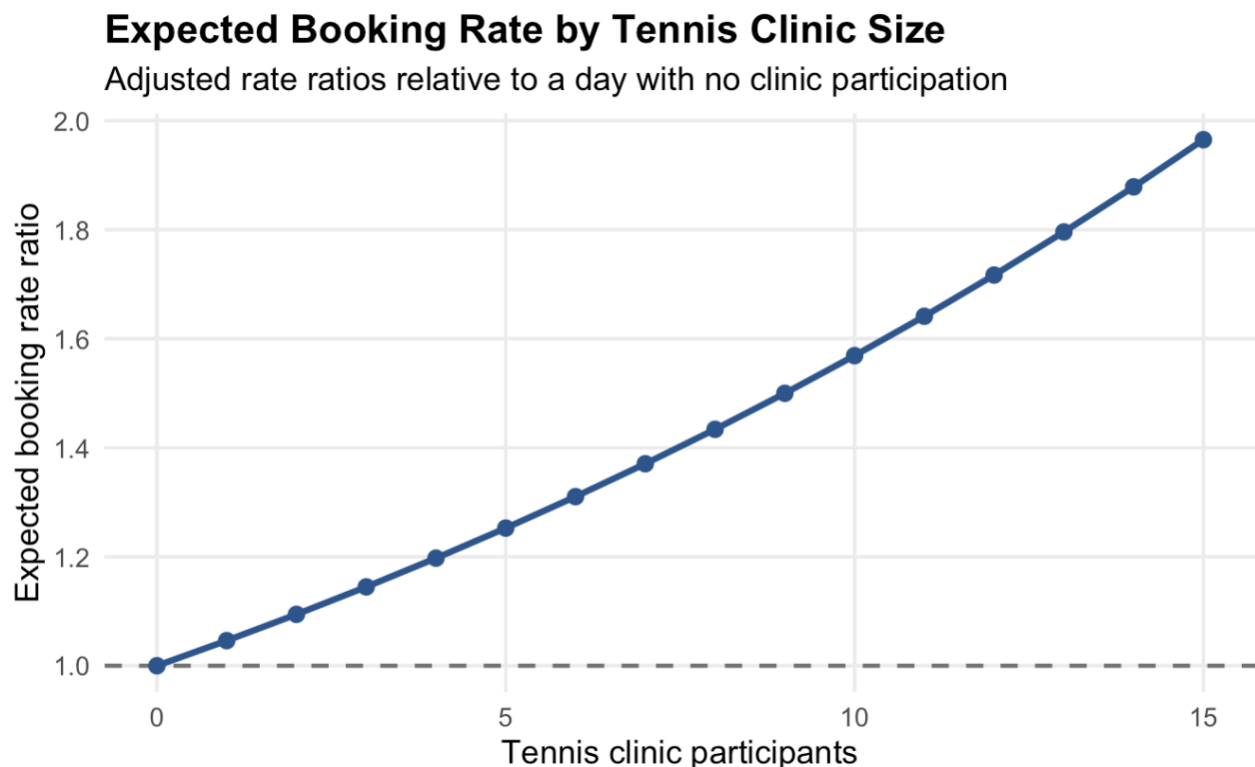
The negative-binomial models did not find evidence that tennis clinics reduced aggregate daily member and inn-guest court-booking volume. Instead, clinic participation was positively associated with regular court bookings after accounting for weekday, month, and year.

In the primary continuous clinic model, each additional clinic participant was associated with an estimated 4.6% increase in the expected number of daily regular court bookings ($p < 0.001$). This estimate is based on a booking rate ratio of approximately 1.05 per additional clinic participant. In other words, days with larger clinics also tended to have higher levels of regular court-booking activity, even when comparing days within the same weekday, month, and year.

This result should be interpreted as an association rather than a causal effect. It does not necessarily mean that clinics directly increase regular bookings. Clinics may be scheduled on days when overall tennis demand is already high, or they may encourage participation by helping less experienced players gain the skills and confidence needed to reserve courts independently.

Below, Figure 1 illustrates this positive association:

Figure 1



To test whether the relationship differed across small, medium, and larger clinics, the second model grouped clinic participation into four categories: no clinic, 1–5 participants, 6–10 participants, and 11 or more participants. No-clinic days served as the comparison group.

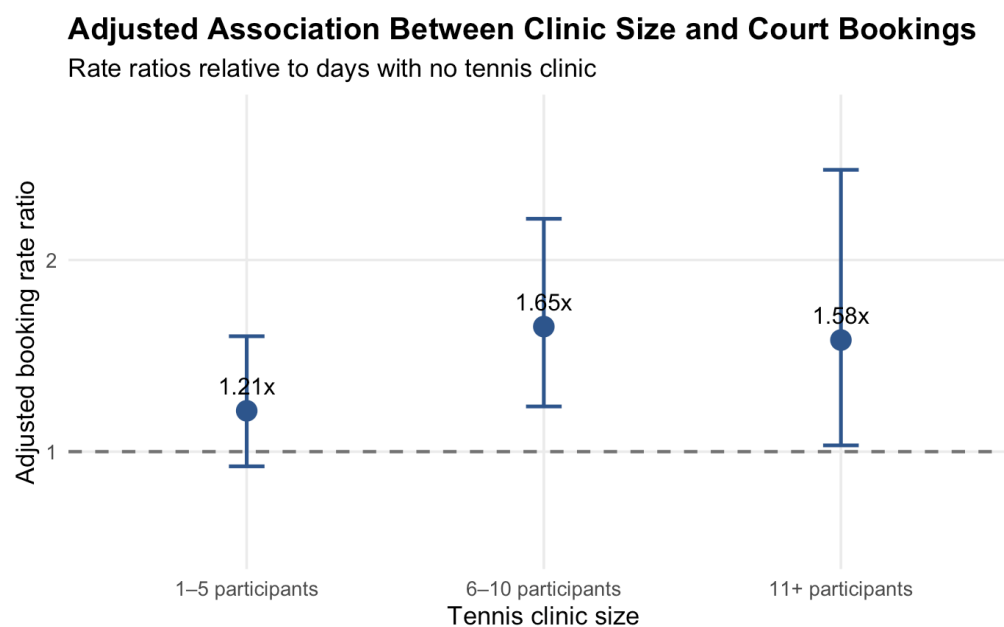
The adjusted association between clinic size and court bookings is shown in Table 2.

Table 2

Clinic size	Rate ratio	95% confidence interval	P-value	Interpretation
No clinic	1.00	Reference	—	Reference group
1–5 participants	1.21	0.92–1.60	0.164	No clear difference from no-clinic days
6–10 participants	1.65	1.24–2.21	< 0.001	65% higher expected booking rate
11+ participants	1.58	1.03–2.47	0.031	58% higher expected booking rate

Figure 2 displays the adjusted booking rate ratios for each clinic-size group. The dashed reference line at 1.00 represents no difference from no-clinic days. The confidence intervals for the 6–10 and 11+ participant groups remain entirely above 1.00, indicating higher expected booking volume than on comparable no-clinic days. The confidence interval for the 1–5 participant group crosses 1.00, indicating that this smaller-clinic estimate was not conclusive.

Figure 2



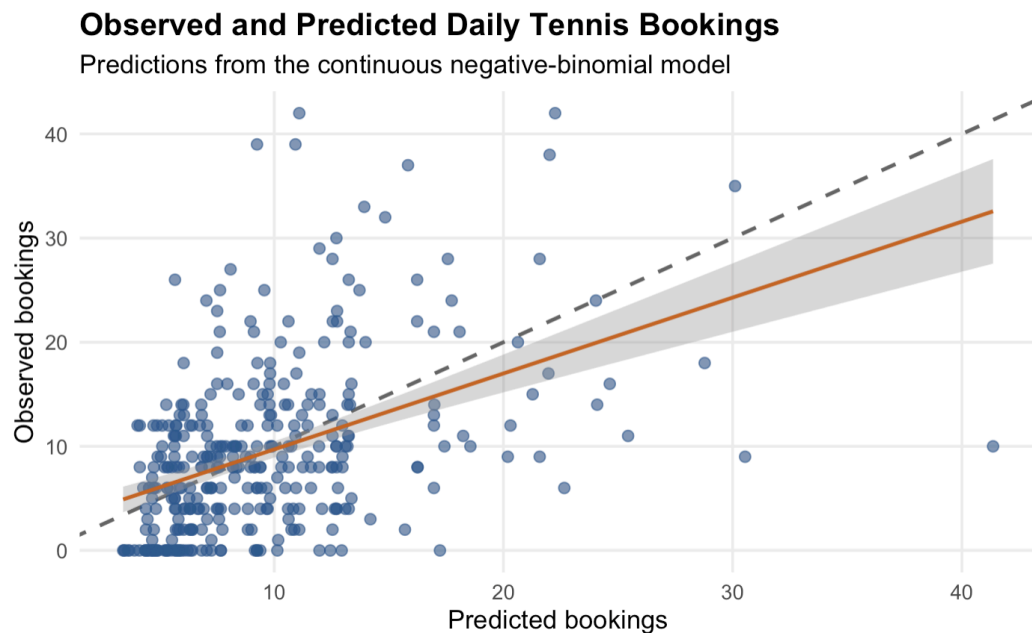
To assess how well the continuous negative-binomial model captured daily booking patterns, predicted booking counts were compared with the observed number of member and inn-guest bookings for all 351 days in the dataset. The model's mean predicted booking count was 9.48 bookings per day, which was very close to the observed average of 9.32 bookings per day. This indicates that the model estimated the overall average level of daily court use reasonably well.

However, predictions for individual days were less precise. The model had a mean absolute error of 5.59 bookings, meaning that its daily prediction was off by about 5.6 bookings on average. The root mean squared error was 7.47 bookings, which gives additional weight to days with especially large prediction errors. Predicted and observed booking counts had a moderate positive correlation of 0.44, indicating that the model generally predicted higher booking volume on days that actually had higher booking volume, although substantial day-to-day variation remained unexplained.

The model predicted the observed booking total within three bookings on 33.6% of days and within five bookings on 57.5% of days. This level of accuracy is reasonable for an operational dataset with only clinic participation and broad calendar variables, but it also shows that other factors likely affect daily court demand. These may include weather, inn occupancy, special events, holidays, court maintenance, and member booking preferences.

In figure 3 (see below), each blue dot represents one day in the dataset, with its horizontal position showing the model's predicted number of bookings and its vertical position showing the actual observed number of bookings. The gray dotted diagonal line represents perfect prediction: points directly on this line would have identical predicted and observed booking counts. Blue points above the dotted line indicate days when the model underpredicted bookings, while points below the line indicate days when the model overpredicted bookings. The orange line summarizes the overall relationship between predicted and observed bookings; its upward slope shows that days predicted to have more bookings generally did have more observed bookings. The gray shaded area surrounding the orange line is the confidence interval, showing the uncertainty around that estimated trend. Although the model captures this general upward pattern, the wide spread of blue points around the dotted line indicates substantial day-to-day variation that the model does not explain. Several high-booking days fall well above the perfect-prediction line, meaning the model substantially underpredicted demand on those days. This reinforces that the model is more useful for evaluating the overall association between clinic activity and booking volume than for making precise forecasts for a particular day.

Figure 3



Conclusion

High Hampton Country Club has a limited number of courts available during the summer peak season, requiring management to balance regular member and inn-guest reservations with organized programming such as tennis clinics. Because clinics can occupy courts that might otherwise be used for regular play, this analysis examined whether clinic participation was associated with lower daily court-booking volume.

Contrary to the expectation that clinics might reduce regular court access, the results showed no evidence of lower aggregate daily member and inn-guest booking volume on clinic days. After accounting for weekday, month, and year, each additional clinic participant was associated with an estimated 4.6% increase in expected daily regular bookings. The category model showed a similar pattern: compared with no-clinic days, days with 6–10 clinic participants had an estimated booking rate 65% higher, and days with 11 or more clinic participants had an estimated booking rate 58% higher.

These findings suggest that tennis clinics do not appear to crowd out regular court use when booking activity is measured across an entire day. Instead, clinic days tended to coincide with greater overall tennis activity. One possible explanation is that clinics are scheduled on days when demand for tennis is already high. Another possibility is that clinics help newer or

less experienced members build the skills and confidence necessary to play independently, which could encourage later court reservations. Clinics may also be held during lower-demand time periods, allowing regular bookings to continue during other parts of the day.

However, this analysis does not prove that clinics directly increase regular bookings or that they impose no access cost on members and inn guests. The model identified an association, not a causal effect. In addition, the available records measure total daily bookings rather than court-hours, reservation times, waitlists, or the number of courts occupied by each clinic. A clinic could still reduce access during a particularly desirable time of day, such as a busy morning or afternoon period, even if total daily booking volume remains high.

The model also captured average booking patterns more effectively than individual daily outcomes. While its average predicted booking count was close to the observed average, daily predictions had a mean absolute error of 5.59 bookings and a moderate correlation of 0.44 with actual bookings. This remaining variation suggests factors not included in the model, such as weather, inn occupancy, special events, holidays, court maintenance, booking time preferences, clinic timing, and court allocation, which likely influence day-to-day court demand.

Future analyses should collect information on the specific start and end times of clinics, the number of courts allocated to each clinic, available court-hours, reservation denials or waitlists, inn occupancy, and clinic revenue and staffing costs. These data would allow the club to determine whether clinics are scheduled during unused capacity, whether they limit access during high-demand time slots, and whether their financial and participation benefits outweigh any reduction in member or inn-guest court availability.

Overall, the evidence from four summer seasons suggests that tennis clinics at High Hampton Country Club are compatible with strong daily regular court-booking activity. Rather than asking only whether clinics should be offered, future planning should focus on when clinics are scheduled, how many courts they use, and how the club can preserve reliable access for members and inn guests during its busiest periods.